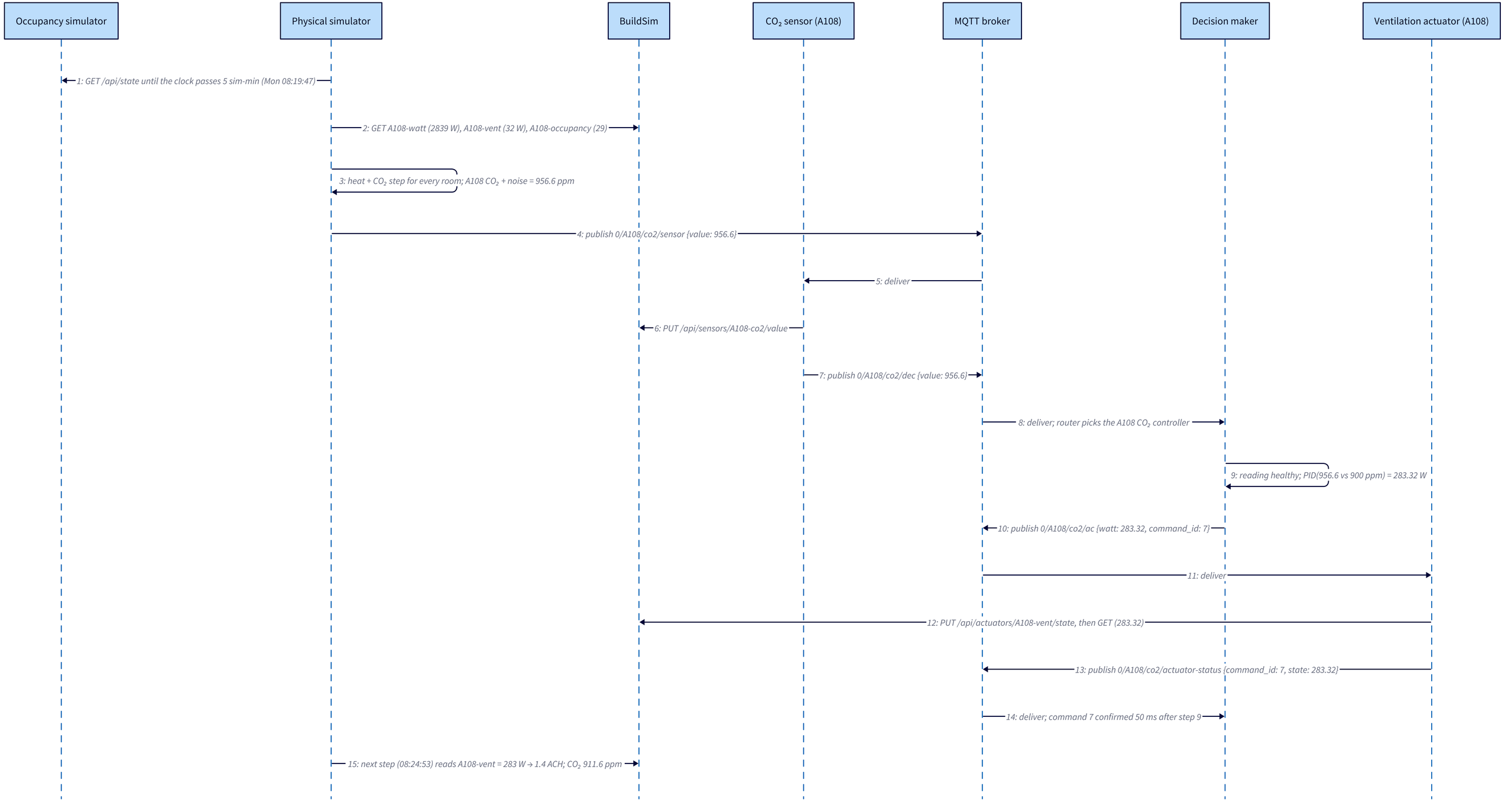

Occupancy simulator
Physical simulator
BuildSim
CO2 sensor (A108)
MQTT broker
Decision maker
Ventilation actuator (A108)
1: GET /api/state until the clock passes 5 sim-min (Mon 08:19:47)
2: GET A108-watt (2839 W), A108-vent (32 W), A108-occupancy (29)
3: heat + CO2 step for every room; A108 CO2 + noise = 956.6 ppm
4: publish 0/A108/co2/sensor {value: 956.6}
5: deliver
6: PUT /api/sensors/A108-co2/value
7: publish 0/A108/co2/dec {value: 956.6}
8: deliver; router picks the A108 CO2 controller
9: reading healthy; PID(956.6 vs 900 ppm) = 283.32 W
10: publish 0/A108/co2/ac {watt: 283.32, command_id: 7}
11: deliver
12: PUT /api/actuators/A108-vent/state, then GET (283.32)
13: publish 0/A108/co2/actuator-status {command_id: 7, state: 283.32}
14: deliver; command 7 confirmed 50 ms after step 9
15: next step (08:24:53) reads A108-vent = 283 W → 1.4 ACH; CO2 911.6 ppm